

Outbreak prediction using AI

Supercomputers running on AI algorithms can help prevent outbreaks like the coronavirus and stop any pandemic before it happens. BlueDot, an automated infectious disease surveillance computer system created by a Canadian AI company, was able to predict coronavirus outbreak in Wuhan city long before it happened. BlueDot had sent early warnings about the outbreak and even predicted spread of the virus to other places like Bangkok, Seoul, Taipei, and Tokyo in the days following its initial appearance.

BlueDot uses machine learning and natural language processing techniques to prepare its model that sifts through news reports in 65 languages, airline ticketing data, and animal disease networks. With the help of AI algorithm, it then makes a meaningful prediction about any infectious outbreak.

Self-driven cars/autonomous vehicles

During the outbreak of coronavirus in Wuhan city, many people tried returning to their hometowns, mainly through flights. They tried hiring taxis and other public transport vehicles to reach airport, but many taxi drivers refused to provide service because of the fear of getting infected. If self-driven autonomous vehicles were available, everyone could get to the airports during the outbreak.

A self-driven car uses a variety of sensors like radar, lidar, sonar, GPS,



Self-driving/autonomous car

odometry and inertial measurement units, along with AI algorithms, to drive from one place to another autonomously. A company called Waymo, initially led by Google, has successfully tested their self-driven cars and is planning to manufacture Waymo Level 4 autonomous vehicles very soon. Uber is also interested in providing self-driven taxi service and might use such self-driven cars for this purpose in the near future.

Delivery drones

When we think of drones, the first thing that comes to our mind is high-altitude video recording/streaming. Because that's what most people are doing with drones at present—to capture stunning

videos for their Vlog. But drones can do more than that, and in many places drones are being used to deliver food and pharmaceutical products also.

A company called Flirtey uses drones to deliver water bottles, food and medicines to rural areas in the United States. Flirtey is the first delivery drone company in the world that has been approved by the Federal Aviation Administration to provide delivery service using drones. Another company called Zipline International uses drones to deliver pharmaceutical products to various parts of Rwanda in Central Africa.

Amazon, the e-commerce tech giant, is also planning to use drones for the delivery of its online orders; the service will be called Amazon Prime Air. Such delivery drones use GPS navigation and AI algorithms to fly from one location to another.

During the coronavirus outbreak in China, people faced a lot of difficulty in buying food and medicines, because most stores remained closed. Using drones to deliver food and medicines during an outbreak can help the people to survive such hardships.

During outbreak of coronavirus infection, the companies who used AI for their products and services were able to help the people of China. This proved that improving the infrastructure by using AI for various products and services can help improve our life, besides controlling or preventing diseases. **EFY**



AI drones for delivery